Generative AI vs. Agentic AI

The evolution of AI can be understood in three stages:

(a) Traditional AI ⇒ It focuses on finding mathematical relationships and patterns between inputs and outputs, primarily for classification or regression tasks (e.g. spam detection).

(b) Generative AI ⇒ It focuses on learning the distribution of data to generate entirely new samples (text, images, audio, code, etc) that mimic human creation. However, generative AI is inherently reactive; it requires step-by-step human prompts and lacks autonomous planning.

(c) Agentic AI ⇒ A behavioural paradigm where the AI is given a role and goal, and it independently plans, acts, adapts, and completes the task with minimal human intervention.

Generative AI is a building block (capability) that Agentic AI uses to achieve its behaviour.

Key Characteristics of Agentic AI

A true Agentic AI system must possess these six traits:

1. Autonomous :- Makes decisions and takes actions without step-by-step instructions. Autonomy can be controlled via boundaries, human-in-the-loop checkpoints, override controls and guardrails.

2. Goal-Oriented :- Operates with a persistent objective in mind, often bound by specific constraints (e.g., budget or location restrictions)

3. Planning :- Breaks high-level goals into a sequence of sub-goals. The agent generates multiple candidate plans, evaluates them based on efficiency, risk and tool availability, and selects the most optimal path.

4. Reasoning :- The cognitive process of interpreting environmental feedback, drawing conclusions, and making decisions during both the planning and execution stages.

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5. Adaptability :- The ability to modify strategies mid-task due to tool failures, unexpected external feedback, or changes to the original goal.

6. Context Awareness :- Uses memory to retain relevant information accross the workflow. Short-term memory tracks current session data (e.g. tool API responses), whilst long-term memory stores overarching goals, past interactions, and user preferences.

Core Components of Agentic AI

At a high level, an agentic architecture relies on five component

1. Brain (LLM) ⇒ The core intelligence that interprets goals, handle reasoning, creates plans, and decides which tools to invoke.

2. Orchestrator ⇒ The "nervous system" that executes the plan, manages task sequencing, conditional routing, loops, and retries. LangGraph is the framework used to build this orchestrator.

3. Tools ⇒ External interfaces (APIs, databases, RAG knowledge bases) that allow the agent to interact with the outside world.

4. Memory ⇒ Stores both short-term session state and long-term user context.

5. Supervisor ⇒ Manages human-in-the-loop interactions, enforces safety guardrails, and escalates complex edge cases to humans.

Workflows Vs. Agents

1) **Workflows** ⇒ Systems where LLMs and tools are orchestrated through predefined, hardcoded paths. The developer decides the sequence of execution.

2) **Agents** ⇒ Systems where LLMs dynamically direct their own processes and tool usage. The agent plans the workflow and decides which step to take and in what order based on the initial prompt.

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